SITUATION MANUAL

Produced by: Exercise Planning & Conduct Team

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1. Exercise Overview

Exercise Name	**Wildfire Risk Modeling Exercise**
Exercise Conduct Period	March 24, 2026 – May 21, 2026
Scope	This exercise is designed around the risk of wildland fire being carried by vegetation into the wildland-urban interface and intermix, and into the built environment impacting people and infrastructure.
Focus Areas	• Mitigation • Response
Primary Core Capabilities	• **Planning:** Conduct a systematic process engaging the whole community as appropriate in the development of executable strategic, operational, and/or tactical-level approaches to meet defined objectives. • **Risk & Disaster Resilience Assessment:** Assess risk and disaster resilience so that decision makers, responders, and community members can take informed action to reduce their entity's risk and increase their resilience. • **Long-Term Vulnerability Reductions:** Build and sustain resilient systems, communities, and critical infrastructure and key resources lifelines so as to reduce their vulnerability to natural, technological, and human-caused threats and hazards by lessening the likelihood, severity, and duration of the adverse consequences.
Objectives	• Identify the unique strengths of all existing wildfire risk models, and their most appropriate use cases. • Determine how existing models can be ensembled to achieve the greatest accuracy and precision. • See below for more detailed objectives.
Threat or Hazard	• Wildland Fire • Wildland Urban Interface (and Intermix) Fire • Urban Conflagration / Structure-to-Structure Fire Spread • Single Structure Fire

Scenario	Two (2) fictitious landscapes with different topography, vegetation, weather, and built communities/infrastructure. The two synthetic/fictitious landscapes are referred to as: • Town of Forests • Town of Prairies Each landscape comprises a 5km X 5km area, defined through real spatial geometries, surface characteristics, and road networks. Synthetic (fictious) spatial data is provided for buildings with characteristic attributes, parcel boundaries tied to buildings, tree data capable of voxelization, surface fuels, point ignitions, and synoptic weather data. *See the section on "Scenarios and Data" for more details*.
Exercise Planning & Conduct Team	Google.org, UL Research Institutes | Fire Safety, Gordon and Betty Moore Foundation, California Department of Forestry & Fire Protection CAL FIRE Since 1885, SERDP, Wildfire Science & Technology Commons, FireWERX
Participating Organizations	~47 organizations with unique wildfire risk-related models, representing academia/research, private sector, and government.
Point of Contact	**General Questions:** FireRisk@ul.org **Wildfire Commons Questions:** info@wildfirecommons.org

2. Background

Wildfire can be beneficial to some ecosystems in certain conditions however, we're currently experiencing an increasing trend in wildfires destroying thousands of buildings, claiming lives, and resulting in negative economic impacts with annual costs estimated in the hundreds of billions of dollars.

Yet we struggle to consistently measure, understand, and predict fire risk - the very foundation of mitigation. In response, there is an influx of science and technology developments in wildfire hazard exposure, risk quantification, and prediction.

Now more than ever, we need to **understand the unique strengths of wildfire risk models and identify how models can be ensembled** to achieve the most precise intelligence on fire risk. To answer this call to action, several organizations came together to cooperatively design and conduct the Wildfire Risk Modeling Exercise.

Key Definitions

1. ***Wildfire*** *is any* ***unplanned and uncontrolled vegetation fire that,*** *regardless of ignition source, may* ***negatively affect social, economic, and/or environmental values,*** *and require suppression response or other action.*

2. ***Risk*** *is the* ***potential consequence*** *which could occur to a system (i.e. landscape, tree, animal species, infrastructure site, etc.), society or a community, often determined probabilistically* ***as a function of hazard, exposure, vulnerability, severity of consequence, and capacity to withstand impact.***

2.1 Goals and Objectives

Goal

- Advance wildfire risk modeling science through interactive exercise and workshop, while fostering collaboration among leading research organizations.

Scope

- The scope of this exercise is inclusive of fire risk models that support natural environments, built communities, and infrastructure.
- Emphasis on fire risk from the wildland, carries by vegetation into the WUI, and propensity for structure-to-structure fire spread.

- Designed as a no-fault forum for all organizations to apply their models in a common simulated environment that encompasses wildland, WUI, and urban environments. This format enables all participating organizations to learn how their models respond to different parameterized variables within natural and built environments and inform future model development.
- All participating organizations will be equally recognized for their scientific and technical merits, resulting in win-win outcomes.

Objectives

- Identify the unique strengths of all existing wildfire risk models, and their most appropriate use cases.
- Determine how existing models can be ensembled to achieve the greatest accuracy and precision.
- Identify common data requirements that, if available, would increase accuracy, reliability, and precision.
- Determine the requirements to scale wildfire risk models, and any technical degradation in scaling.
- Determine the feasibility of establishing a common Wildfire Modeling Interconnected Project as a shared framework for model standardization and interoperability.
- Identify opportunities for open science, code, and methodology frameworks that can be shared for innovation.
- Identify priority gaps in wildfire risk modeling science that can inform future scientific advancements and investments.

2.2 Desired Outcomes and Benefits

For Participants & Organizations	For Community
Learn how your model responds to different parameterized variables within natural and built environments.	Identify specific strengths of existing models, and how they can best be ensembled.
Use the exercise as a model innovation accelerator for your research & development efforts.	Gain independent understanding of the appropriate uses of existing models.
Gain actionable insights to inform future model development.	Connect practitioners with model developers/researchers for co-development opportunities.
Contribute to advancing wildfire risk modeling science.	Identify and prioritize gaps in modeling science, core data resources, or other related areas.

Build relationships and collaboration among science, technology, and philanthropy leaders.	Generate findings that inform future investments in wildfire risk scientific discovery.

2.3 Eligibility Criteria

1. Any organization that has developed a wildfire hazard exposure, vulnerability, and/or risk model.
2. Eligible models must address one or more of the following types of fire: wildland, WUI, structure-to-structure or urban conflagration, and single structure.
3. Eligible organizations include, but are not limited to, academic/research institutions, private companies, non-profit organizations, and public sector entities that have existing or emerging wildfire risk models.
4. Organizations with a wide range of wildfire risk models can participate, this includes both static and dynamic models:
    a. Static Models: Typically, statistical/empirical-based models that provide a snapshot of relative risk. May (or may not) have been developed using dynamic methods and simulation repetitions.
    b. Dynamic Models: Often physics, reduced-physics, or semi-physics based and are time-resolved fire progressions that are run for scenario-specific outcomes.

Keep in mind the scope of the exercise is inclusive of fire risk models that support natural environments, built communities, and infrastructure. There is an emphasis on wildfire risk modeling that conveys the likelihood and severity of wildfire impacts starting in the wildland, carried by vegetation into the WUI, and built environments.

2.4 Exercise Framework

What it IS About	What it's NOT About
✓ Open forum, all organizations can participate.	X No limitations on who can participate.
✓ Encourage collaborative engagement among organizations & teams.	X Not competitive, no winners or losers.
✓ No-fault environment, recognizing the unique merits of all models.	X No consensus-based discussions.
✓ All engagements are listening forums to share individual insights.	X No compensation or other incentives for participation.

✓ Purpose is for scientific learning and discovery.	X No sharing or submissions of information subject to intellectual property rights, non-disclosures, or other limitations.

3. Exercise Schedule and Timeline

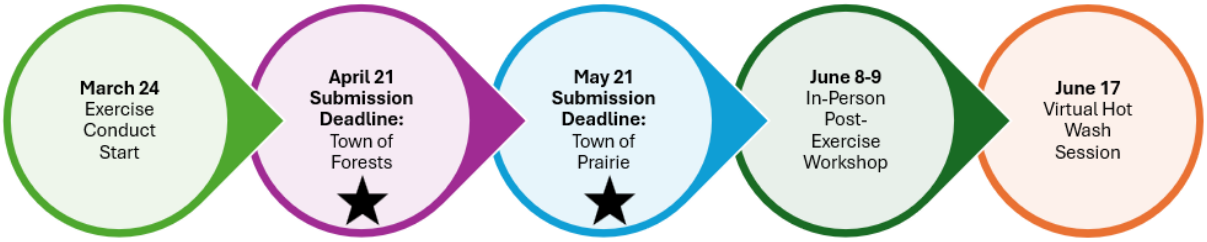
3.1 Key Milestones and Deadlines

Date (all 2026)	Milestone Description
Jan 21 – Mar 9	Open Call for Participation and Registration Open
March 9	Registration Closed
March 18	**Kick-Off Meeting with Participating Organizations**
March 18-24	Onboarding all Participating Organizations onto the Wildfire Commons
March 24	**START Exercise:** Exercise Data Available in Wildfire Commons for all Participants
March 31	1st Virtual Check-In Mtg with Participants
April 14	2nd Virtual Check-In Mtg with Participants
April 21	**Submission Deadline Module A – Town of Forests**
April 23	3rd Virtual Check-In Mtg with Participants
May 7	4th Virtual Check-In Mtg with Participants
May 21	**Submission Deadline Module B – Town of Prairies**
June 8-9	**In-Person Post-Exercise Workshop** • Location: Palo Alto, CA • One (1) representative per model organization • Will require separate registration, which is forthcoming
June 17	**Virtual Exercise Hotwash** • All participants from all organizations • Evaluators and Observers
Early July	**After-Action Review & Report Released**

Key notes regarding milestone dates and deadlines:

- Exact dates are subject to change.
- Calendar invitations will be provided for all virtual meetings.
- Participating organizations are kindly requested and strongly encouraged to submit their model outputs and documentation by the deadlines indicated above. This allows for completion of evaluations within the exercise timeline.

3.2 Timeline Summary



4. Conduct Rules / Rules of Play

4.1 Participant Roles

The term *participant* encompasses many groups of people and organizations, not just those playing in the exercise. Groups of participants involved in the exercise, and their respective roles and responsibilities, are as follows:

Players: Personnel with organizations that registered their model to participate in the exercise. Players have the most active role in using the synthetic data with their models in conducting the exercise in alignment with the defined scenarios.

Observers: Do not directly participate in the exercise. However, they may support exercise conduct and post-exercise discussions by asking relevant questions or providing subject matter expertise.

Facilitators: Primarily from the Exercise Planning & Conduct team. They provide updates, guidance, technical assistance, and moderate discussions. They also provide additional information or resolve questions as required.

Evaluators: Are assigned to observe and document their observations based on SME reviews of the model submissions in alignment with the exercise objectives.

4.2 Rules of Play

1. This exercise is an open, no-fault forum wherein models will be reviewed in alignment with the exercise objectives.
2. Throughout the various exercise meetings and discussions, varying viewpoints, even disagreements, are expected and should be embraced by all. This is a learning experience and opportunity for all.
3. Run your models using the synthetic Respond to the scenario using your knowledge of current plans and capabilities (i.e., you may use only existing assets) and insights derived from your training.

4. Discussions are not precedent setting and may not reflect your organization's position or model capabilities. This exercise is an opportunity to discuss and present multiple approaches, options, and innovate on possible solutions.
5. Issue identification is helpful! But not as valuable as suggestions and recommended actions that could improve wildfire risk modeling, exercise design, and related efforts. Problem-solving efforts should be the focus.
6. The assumption is that the exercise scenarios are plausible, as are the synthetic exercise data. All players will receive information at the same time.
7. Players only share and submit information as a part of the exercise that is considered open, is not subject to non-disclosure agreements and not otherwise considered protected intellectual property.
8. Participation in this exercise is voluntary and that my organization, company, or affiliated representatives will not receive compensation of any form for participating in this exercise.
9. Full and complete participation in the exercise is encouraged; there is no penalty if an organization is unable to complete participation. If you're unable to complete the exercise, please notify the team at FireRisk@ul.org.
10. Participation in this exercise is focused on underlying modeling science and that no individual person is considered a research subject.

5. Scenarios and Data

5.1 Scenario Overview

- Exercise scenarios are provided for two (2) selected locations with different topography and vegetation.
  - Each location encompasses a small area of wildland with an area in the WUI with vegetation capable of carrying fire, and an urban environment comprised of multiple residential subdivisions and some commercial properties.
  - Exercise locations are not sufficiently large for some wildland fire risk models to run, such as those that are designed for large-scale fires.
  - Exercise locations are designed for models to run in areas where fire in the natural landscape propagates into the WUI and threatening communities and the built infrastructure.
  - Each location is 5 kilometers by 5 kilometers, defined through real spatial geometries, surface characteristics, and road networks.
  - Synthetic (fictious) spatial data is provided for buildings with characteristic attributes, parcel boundaries tied to buildings, tree data capable of voxelization, surface fuels, point ignitions, and synoptic weather data. *See the section on Data for more details*.

Module A: *Town of Forests*
- Mountainous with variable elevation.
- High density of trees with varying species, heights, diameters, etc.
- Some residential sub-divisions are wildland-urban intermix.

Module B: *Town of Prairies*
- Mostly flat, with consistent elevation.
- Lower density of trees and tree characteristics.
- Some residential sub-divisions are wildland-urban interface.

- Each of the two locations encompass an area of approximately 5km X 5km, and varying levels of resolution, including detailed building-level attributes at different parameters, are provided in the semi-synthetic data to allow for flexibility in modeling capabilities.

- Exercise data is made available to all registered organizations and will be accessible through the Wildfire Science & Technology Commons.
  Access to Town of Forests data: https://scil-data.sdsc.edu/data/ul-risk-modeling-2026/Forest/
  Access to Town of Prairies data: https://scil-data.sdsc.edu/data/ul-risk-modeling-2026/Prairie/

- Participants run their models applying relevant information provided, synthetic data (also provided), and submit a suite of model outputs and input documentation, which is detailed in the Model Submission Guidelines to be provided to all participants at the start of the exercise.

- Model outputs are then reviewed in alignment with the objectives outlined above, serving as a common foundation for model science innovation and advancement. The findings from this exercise may be used to inform future investments in wildfire risk scientific discovery through future prize challenges and other opportunities.
- Models are not evaluated or compared to a specific baseline model or other models. Each model is reviewed for its unique strengths and sensitivity to parameterized variables.

5.2 Synthetic Exercise Data

The most important component of the exercise is the synthetic (or fictitious) data sets produced to support this exercise. Synthetic data was produced to ensure that the communities are anonymized for the purposes of the exercise. The synthetic data also achieves standardization in the core, foundational data inputs that models use, thereby providing a baseline from which sensitivity to specific, parameterized variables can be observed. Provided below are several key points regarding the synthetic exercise data:

- Synthetic exercise data is made available to all participating organizations through the Wildfire Commons. *See section below on "Using the Wildfire Commons".*
- Participants are advised to use the synthetic exercise data provided. Please do not replace or modify the synthetic exercise data with an alternative data source.
- Additional data may be used as supplemental data to enhance and support your model.
- Not all models require use of all synthetic exercise data, which is perfectly acceptable.
- Use only data relevant to your models, that your models are sensitive to.

5.3 Summary of Synthetic Exercise Data

- All Raster data files and .Geojson are in WGS84; epsg:4326 unless otherwise stated.
- The elevation Rasters have no data values defined as -3.402823e+38, and all other rasters have no data values defined as 9999.
- To create 3D canopy data, you can re-project <city name>_Treelist.geojson/.txt into an appropriate crs, convert coordinates to absolute coordinates, and use LANL Trees https://github.com/lanl/Trees.

Each folder contains the following data, note that the exact file names and details are subject to change however, this serves as a general guide for the synthetic exercise data provided for each of the scenarios:

- `All rasters and geojsons are in WGS84 (EPSG:4326) unless otherwise stated.`
- `All rasters have no data values defined as 9999.`

- <city name>_depth.tif : 1 band raster file of surface fuel depth [meters]
- <city name>_elevation.tif : 1 band raster file of elevation from USGS DEM products
- <city name>_generated_buildings_fireprops.geojson : geojson containing synthetic building polygons from a separate collaborator
- <city name>_ignition.geojson: geojson containing the ignition location and start time.
- <city name>_moist.tif : 1 band raster file of surface fuel moisture [dec]
- <city name>_roads.geojson : geojson containing road network
- <city name>_rhof1.tif : 1 band raster file of 1 hour surface fuel loading [kg/m^2]
- <city name>_rhof10.tif : 1 band raster file of 10 hour surface fuel loading [kg/m^2]
- <city name>_rhof100.tif : 1 band raster file of 100 hour surface fuel loading [kg/m^2]
- <city name>_SAV.tif : 1 band raster file of fuel surface area to volume [1/m]
- <city name>_SB40.tif : 1 band raster file of Scott & Bugran 40 fuel model [int]
- <city name>_Synoptic_Weather_Data.csv : CSV containing weather observations including:

  timestamp

  air temperature (F)

  relative humidity (%)

  wind speed (mph)

  wind direction (deg)

- <city name>_Treelist.geojson : geojson containing tree data including:

  lat/lon, height (HT) [m],

  canopy base height (CBH) [m],

  maximum canopy diameter (DIA) [m],

  height to maximum canopy diameter (HT_TO_DIA) [m],

  canopy bulk density (CBD) [kg/m^3],

```
canopy moisture (MOIST) [dec],

fine fuel size scale (SS)
```

- `<city name>_Treelist.txt : text file containing the same data as <city name>_Treelist.geojson in txt format`
- `<city name>_ReadDatFiles.py : python script to read and plot the dat files`

```
treesfueldepth.dat : depth

treesmoist.dat : moistures

treesrhof.dat : density

treesss.dat : size and scale
```

5.4 Supplemental Data

A list of potentially relevant, publicly available data for other features are included in Appendix B. This is not intended to be all encompassing but rather serves as a starting point for some of the supplemental data that players may want to incorporate depending on the capabilities and relevance to their unique models. Examples of infrastructure data that may be added supplementally include wind turbine infrastructure, US electrical transmission lines, water resources/reservoirs, railroad networks, etc.

Participants are ***not*** expected to use supplemental data but are welcome to innovate as long as the core synthetic exercise data are used where applicable by a model.

6. Model Submission Guidelines

6.1 Submission Requirements

1. **Model Inputs**
   a. **Core Inputs: Synthetic Exercise Scenario Data**

The required Synthetic (fictitious) Exercise Scenario Data inputs are defined in the Dataset Details section in FireForge for this exercise and are accessible in the Wildfire Commons workspaces provided to each participating organization. These synthetic datasets are the core inputs and are expected to be used as they are provided (with provided values and attributes unchanged) as model inputs to the extent possible.

   b. **Additional Model Inputs**

If a team generates additional datasets/derived datasets (features, masks, intermediate outputs, extracts, etc.) that are model-relevant inputs, they should be added to your organization's workspace with:

- A clear access point (fetch URL(s), API endpoint(s), or other resource links)
- Sufficient metadata to interpret the dataset (format, schema/bands, CRS, extent, version/date)

- Authentication/access requirements (if applicable). Access-controlled inputs are allowed, but please provide information on how access is granted.

When transformations are applied to generate model-relevant inputs from the core inputs provided in the exercise scenario data, participating organizations are encouraged to provide a git repo demonstrating the processing steps end-to-end.

2. **Model Outputs**
    a. **Output Files:**

All participating organizations must deliver their output files in non-proprietary formats, and are highly encouraged to submit outputs as:

- Raster: GeoTIFF
- Vector: GeoJSON, Shapefiles (.shp)

All outputs must include appropriate metadata and descriptive files so that evaluators can understand and properly interpret their outputs.

b. **Plus, Esri-compatible Map Services or Feature Services**

Additionally, participating organizations are requested to also provide model outputs as Esri-compatible Map Services or Feature Services. These versions will be used to support evaluators in consistently reviewing model outputs as basic spatial visualizations, recognizing that this is one component of the evaluation process. Instructions are as follows:

- If your organization already uses ArcGIS Online, you'll provide your username and be granted access to the dedicated exercise "Group" where you will share your Esri-compatible Map Services or Feature Services. There are secured "Groups" only accessible to exercise participants and evaluators.
    - Note: Be sure to set-up your Esri-compatible Map Services or Feature Services to be "view only" and not editable.
- If your organization does not already use ArcGIS Online, you'll be provided with a temporary user account to create your feature services and granted access into a dedicated exercise "Group" where you will share your Esri-compatible Map Services or Feature Services.
- If your organization does not have an ArcGIS account, please contact FireRisk@ul.org, so a temporary account can be established.
- If your organization does not have the skills or capacity to create Esri-compatible Map Services or Feature Services, please contact FireRisk@ul.org. Technical assistance can be provided to convert your raster and/or vector outputs into compatible formats.

c. **Supporting Jupyter Notebooks *(When Applicable)***

Participants are encouraged to provide a git repo with one or more reproducible Jupyter Notebooks demonstrating the model workflow from end-to-end. Notebooks should include:

- A clear step-by-step execution flow
- Input dataset references (source + version/date where applicable)
- Output paths and naming conventions
- Parameters and assumptions (thresholds, CRS, resampling, tiling scheme, etc.)
- Saved outputs from notebook execution (when relevant/required)

Providing notebooks is especially relevant when:

- The model is hosted in a repository or local package
- The model is programmatically accessible via an API or SDK
- The workflow involves multiple preprocessing, feature engineering, or postprocessing steps

d. Log Files *(When Applicable)*

If the model cannot be executed via a Jupyter Notebook (e.g., proprietary desktop software or external systems), participants can include a log file or execution record demonstrating the successful execution of the model as well as relevant run configuration details and runtime context (as applicable). Participants must also summarize any important execution details in the final report or supporting documentation.

3. Final Report (PDF)

Each participating organization must submit a PDF report documenting their full workflow and results. The report must include:

a. Workflow Diagram + Full Workflow Description

Include a workflow diagram and narrative description covering the full pipeline end-to-end, including (as applicable):

- Data access and extraction
- Data transformations and processing
- Feature engineering / intermediate products
- Modeling and inference steps
- Post-processing and output generation

b. Model Architecture Diagram + Explanation

Include an architecture diagram and explanation that clearly communicates:

- The model type and major components
- Inputs, outputs, and internal stages
- Training vs. inference flow (if applicable)
- Hardware requirements (CPU, Memory, GPU)
- Key design choices relevant to performance and reproducibility

c. Inputs Used (Core + Derived / Model-Relevant)

The report must list all inputs used by the model, including core inputs (Exercise Scenario Data) and derived/model-relevant inputs created by the participating organization. For every non-core input, participant must document:

- What it is (name, description, format)
    - Indicate how specific inputs were used at the attribute level (i.e. roof type: non-combustible roof vs. wood roof)
- Spatial/temporal resolution (where applicable)
- CRS, extent, schema/bands (where applicable)
- Key metadata and version/date
- How it was created (transformations, filtering, resampling, joins, feature engineering)
    - What assumptions did you make in your models about the data elements and attributes, and indicate if you had to augment any attributes to achieve the sensitivities that your model requires
- Where it is used in the workflow

d. Model Outputs

Clearly identify the primary outputs generated by the workflow. For each, include:

- What it represents and brief guidance on how the output should be interpreted that you'd deliver to the intended end user for their use.
- Format and data type
- Spatial/temporal resolution and CRS (if applicable)
- Units, value range, or classification scheme

e. Downstream Use of Model

Please explain how you imagine that model outputs will be used in real-world contexts. Please specify the intended use of your model if you've determined that at this point. Examples may include, but are not limited to:

- Risk Assessment for (local) Community Wildfire Protection Planning
- Community Risk Reduction Planning and Resource Allocation
- Wildfire Mitigation (i.e. fuels reductions: mastication, prescribed burning, defensible space, structural hardening, etc.) Planning, Prioritization, and Resource Allocation
- Response Pre-Incident Planning and Resource Allocation Planning
- Risk Conditions Monitoring for Readiness and Incident Size Up
- Response Resource Planning during Initial Attack
- Response Resource Management throughout Response Operations

f. Assumptions, Parameters, and Validation

The report can also include:

- Key assumptions
- Parameter choices (thresholds, hyperparameters, tiling schemes, etc.)
- Validation approach (if applicable), including what was evaluated and how

g. Additional Supporting Documentation (When Applicable)

Include any supporting files required to understand or reproduce the work, such as:

- README files
- Configuration files

- Model documentation
- Model-relevant publications or references

Additional Guidance for Model Submissions

To help elucidate the instructions above, a mock submission will be provided in the Wildfire Commons that includes the following:

1. **Model inputs**
   a. Additional model inputs (anything extra you need for your workflow beyond what is provided in the scenario data)
   b. Git repo (when applicable) with notebooks used to produce model inputs

2. **Model outputs**
   a. Raster and/or vector files in non-proprietary formats
   b. An Esri-compatible Map service or feature service
   c. Git repo with notebooks used to produce model outputs

3. **PDF of final report**

6.2 Mock Submission Package

Within the dedicated data challenge site for the Wildfire Risk Modeling Exercise within the Wildfire Commons is a "Mock Submission Package". The purpose of the Mock Submission Package is to provide participants with a mock submission that is not representative of a wildfire risk model however, it illustrates the submission components and example content for a complete submission. This example was produced using open-sourced code; thus it's an example of a fully reproducible model. We recognize that not all participants are using open-sourced code and therefore your model submissions will vary accordingly, modifications for this are reflected above in section 6.1.

The model submission package contains one IPython notebook that uses the scenario data provided to create fuel and landscape layers that can be input into a FARSITE simulation. The FARSITE simulation is defined using the provided parameters (location, weather, landscape also from the scenario data), ignites a 5-hr FARSITE simulation, and retrieves the FARSITE perimeter predictions. The perimeters are overlaid on the scenario building data provided to calculate structures at risk.

7. Communication & Technical Assistance

Check-In Sessions (Virtual)

A series of 1-hour check-in sessions will be hosted throughout the exercise conduct period. The purpose is to check-in on exercise conduct, progress on each module, address key

questions, and provide general updates. These will be scheduled with calendar invitations and all player/participants will be invited. Participation in the check-in sessions is highly encouraged.

Office Hours for Technical Assistance

For players/participants that would like technical assistance during exercise conduct weekly office hours have been established. You can schedule a technical assistance session for this exercise here: https://calendly.com/pramonettivega-ucsd/fire-risk-modeling-technical-assistance.

Technical assistance is offered to help guide users through the Wildfire Commons, how to set-up data pipelines using the synthetic exercise data, walk through executable notebook development, and support with the model submission process.

General Questions and Help

- General Questions: FireRisk@ul.org
- Wildfire Commons Questions: info@wildfirecommons.org

8. Evaluation Process

Players will submit their model outputs & documentation by the deadline for each Module. Evaluators will review the model submissions for the models they've been assigned. There will be multiple evaluators reviewing each model submission, both Science/Technical Evaluators – and – Practitioner Evaluators. Evaluators will document their reviews using a web-based form that it provided, which aligns with the respective evaluation rubric. ***There are no right or wrong answers, and there is no "scoring" of submissions as part of the evaluation.***

The evaluation criteria are built out based on the exercise objectives, scenario details (when relevant to a model), and the specific parameters in the semi-synthetic exercise data (topography, vegetation, individual building characteristics, mitigations, etc.). Further, not all exercise objectives, scenario details, or exercise data attributes will be relevant to all models – which is perfectly acceptable.

Also, evaluation criteria inherently differ between static vs. dynamic models. Static models are largely statistical and don't account for scenario-specific information that is temporal in nature (i.e. specific weather conditions at a certain date and time). Dynamic models are those that require specific scenario inputs to generate a result. The evaluation framework and process support these inherent model variations.

Example of Evaluation Criteria

Provided below is a small snapshot of several example exercise evaluation criteria and example response values.

EXAMPLE - Evaluator Question	Response Option 1	Response Option 2	Response Option 3
Does the model output provide a rank, score, or value for "Risk"?	Yes	No	Somewhat
If YES - Indicate the type of risk ranking or score provided.	Binary	Numerical	Categorical
Which of the following building variables does the model account for? Select all that apply.	Structural Separation Distance	Year Built	Proximity to Fire Prone Area
Did the model account for topographic features in the Surface data? And to what extent? Select one and provide notes in the narrative.	Yes	No	Somewhat
Did the model apply the synoptic weather data provided?	Yes	No	Somewhat

9. Post-Exercise Workshops

In-Person

An in-person Post-Exercise Workshop will be conducted June 8-9 in Palo Alto, CA. Each modeling organization will be invited to have one (1) representative for their organization to participate in the workshop. Additionally, members of the Exercise Planning & Conduct Team, Evaluators, and select Observers will be invited to participate.

The purpose of the in-person Post-Exercise Workshop is to conduct a collaborative after-action review on exercise conduct, identify findings identified, discuss opportunities for improvement, discuss critical gaps in wildfire risk modeling science, and begin to develop a strategy to advance the science.

Additional information, with a separate registration, for the in-person workshop is forthcoming in April however, the dates and location for the in-person workshop are firm.

Virtual

A virtual post-exercise Hot Wash Session will be conducted for all model organization team members to participate, in addition to the Exercise Planning & Conduct Team, Evaluators, and select Observers. The purpose of this virtual Hot Wash Session will be to conduct a

smaller-scale version of the in-person workshop but with ALL players, and in a virtual format.

10. Using the Wildfire Commons

The Wildfire Science & Technology Commons (Wildfire Commons) hosted by the University of California, San Diego is the primary exercise conduct and collaboration platform that will be used throughout the exercise. A dedicated "site" for the exercise has been established in the Commons, which all model organizations will use to access exercise information, access synthetic exercise data, collaborate with their team during exercise conduct (or among organizations when appropriate), and submit their model outputs and documentation.

- **Link to Wildfire Commons: https://www.wildfirecommons.org/**
- **Video Tutorial for Onboarding onto the Commons:** https://drive.google.com/file/d/1sW8Ti75C91pAvolAuIuNnul-RSm8cVSJ/view

The Wildfire Risk Modeling Exercise is being conducted as a "Data Challenge" within the Wildfire Commons. For basic instructions on how to "Join the Data Challenge" for this exercise, visit: https://documentation.wildfirecommons.org/data-challenges/participating-in-data-challenge/.

- All registered and confirmed organizations have been "whitelisted" into the dedicated data challenge site for the Wildfire Risk Modeling Exercise.
- The primary point of contact (POC) submitted at the time of registration for the modeling organization will serve as the primary contact for their organization within the Wildfire Commons.

Logging into the Commons to Join the Exercise

- We encourage participants to **sign in with the same email they registered with for the exercise.**
- **Institutional credentials are preferred**, with ORCID and Google (Gmail) as alternatives (not preferred).
- If you use a different email to create their account, you will run into a limitation to accessing compute resources.
- If you run into any issues, send an email to: info@wildfirecommons.org.

11. Frequently Asked Questions

1. How do I determine whether my organization or company is eligible to participate?
    a. For eligibility information, review the section above on eligibility criteria.

2. When is exercise conduct?
    a. For dates on exercise conduct, review the section above on key dates.

3. How can I ensure my organization's intellectual property is protected?
    a. Participating organizations are permitted to submit only model output information that *does not contain* information subject to non-disclosure agreements or intellectual property protections.

4. How are participating organizations evaluated?
    a. All modeling outputs will be reviewed by a team of experts. Reviews will be conducted to understand alignment with the exercise objectives outlined above. Participating organizations and their modeling outputs are *not* competitively evaluated or scored. All model organizations, and their models, will be acknowledged for their unique merits.

5. What types of models do this exercise support?
    a. This exercise focuses on wildfire risk models. Models may provide deterministic, probability, likelihood, or exposure of any type of wildfire (i.e. wildland, WUI, urban conflagration). Models at any level of spatial extent and resolution can participate, which you identify as part of the registration. The synthetic data and scenes are developed to allow for flexibility in supporting a diversity of wildfire risk model types.

6. Can organizations participate as teams with another organization(s)?
    a. Yes, organizations can participate as teams in the exercise. However, each organization needs to register separately and should indicate if they are working as a team with another organization.

7. Are there any compensation or honorariums associated with participating in this exercise?
    a. There are no financial compensation or other incentives provided to any participating organization, company, or affiliated representatives.

Appendix A - Glossary of Terms

Term	Definition
Agentic AI	Autonomous system that perceives its environment, makes independent decisions, plans, and takes actions to achieve complex, goal-oriented tasks with minimal human intervention.
AI Agent	Autonomous system that perceives its environment, reasons, plans, and takes actions to achieve complex goals with minimal human oversight, often using LLMs and other AI tools to learn, adapt, and interact with users and systems to solve problems, automate tasks, and even collaborate with other agents.
Binary	Binary models classify areas as either "high risk" or not. The risk score indicates whether an area exceeds a threshold.
Burn Probability	Probability of any location burning due to a wildfire.
Categorical	Categorical models assign discrete risk categories to areas. The risk score is a numeric identifier that maps to a category.
Consequence (or Severity Factor)	Degree of negative impact or seriousness of the outcomes from an event involving one or more hazards.
Dynamic	Often physics, reduced-physics, or semi-physics based and are time-resolved fire progressions that are run for scenario-specific outcomes.
Empirical (or Statistical)	Mathematical model based on empirical data or approximations (i.e. actuarial stats).
Hazard Exposure	Degree to which people, natural capital, buildings, and infrastructure are subjected to potential hazards.
Likelihood of Occurrence	Probability that a given hazard is capable of exploiting a given vulnerability or a set of vulnerabilities.
ML / LLM	Trained algorithm that learns patterns from data to make predictions, classifications, or decisions on new, unseen data.
Numerical	Numerical models provide continuous risk scores that can be visualized in various ways such as quantized buckets, even buckets, custom buckets.
Reduced Physics	Simplifies a complex physical system's high-fidelity description into a computationally efficient, lower-dimensional approximation that captures essential behavior (i.e. reduced order model).
Retrieval Augmented Model	Framework that enhances LLMs by giving them access to external, authoritative knowledge bases (like databases or documents) to retrieve relevant, up-to-date information before generating a response.
Risk	Potential consequences which could occur to a system, society or a community, determined probabilistically as a function of hazard,

	exposure, vulnerability, severity of consequence, and capacity to withstand impact.
Semi Physics	Mathematical model based partly on theoretical principles and partly on empirical data or approximations.
Static	Typically, statistical/empirical-based models that provide a snapshot of relative risk. May (or may not) have been developed using dynamic methods and simulation repetitions.
Urban Conflagration	Building-to-building fire involving multiple structures and/or spreading flame over a large area within a community, typically observed in high density communities.
Vulnerability	Extent of conditions which increase susceptibility to the impact of a potential hazard.
Wildfire	Any unplanned and uncontrolled vegetation fire that, regardless of ignition source, may negatively affect social, economic or environmental values, and require suppression response or other action.[1]
Wildland Fire	Any non-structure fire that occurs in vegetation or natural fuels. Includes wildfires and prescribed fires.[2]
Wildland Urban Interface	Fire in high-density development adjacent to undeveloped wildland vegetation.[3]
Wildland Urban Intermix	Fire in lower-density housing mingled with undeveloped wildland vegetation.[4]

[1] United Nations Office of Disaster Risk Reduction. "Sendai Framework Terminology on Disaster Risk Reduction" < https://www.undrr.org/understanding-disaster-risk/terminology/hips/en0205 >
[2] National Wildland Fire Coordinating Group. "PMS 205: Wildland Fire Glossary of Terms". < https://www.nwcg.gov/publications/pms205/nwcg-glossary-of-wildland-fire-pms-205 >
[3] National Institute of Standards & Technology. "WUI Definitions". < https://www.nist.gov/el/fire-research-division-73300/wildland-urban-interface-fire-73305/hazard-mitigation-methodology-9 >
[4] IBID

Appendix B – Example Supplemental Data Sources

Data Type	Description	Source Link
Community Resilience Estimates	Track how socially vulnerable every neighborhood in the United States is to the impacts of a disaster.	https://www.census.gov/data/developers/data-sets/community-resilience-estimates.html
Community Water System Boundaries	National geospatial dataset of water system service area boundaries for both community and non-community water systems.	https://www.epa.gov/ground-water-and-drinking-water/public-water-system-service-areas?tab=resources
California Interagency Treatment Data	Provides information spatially the location, scope, and extent of fuels treatments completed throughout California.	https://www.wildfirecommons.org/dataset/its-interagency-treatment-dashboard-activity-report-version-2 Feature Service: https://tinyurl.com/eck65h3t
Treatment and Wildfire Interagency Geodatabase	Geodatabase of land management activities related to wildland fire and fuels management. TWIG primarily includes records of management activities intended to affect fuels and modify wildfire outcomes (treatments).	https://reshapewildfire.org/resources/twig-data-resources
U.S. Electric Power Transmission Lines	Depicts electric power transmission lines in the United States. Transmission Lines are the system of structures, wires, insulators and associated hardware that carry electric energy from one point to another in an electric power system.	https://tinyurl.com/mry6k592
EPA Facility Registry Service	Provided in different formats for download and analysis. These files are a part of the EPA Geospatial Data Access Project provides downloadable files of these facilities or sites	https://www.epa.gov/frs/geospatial-data-download-service
U.S. Wind Turbines	Provides the locations of land-based and offshore wind turbines in the United States, corresponding wind project information, and turbine technical specifications.	https://energy.usgs.gov/uswtdb/